Model Comparison Report

California Housing Price Prediction

Maincrafts Internship Task 2

1 Objective

This task extends Task 1 by applying feature scaling, training multiple regression algorithms, and comparing their performance to select the best model for predicting median house value on the California Housing dataset.

2 Methodology

- **Preprocessing**: Features scaled using `Standard Scaler` to ensure fair learning across features of different magnitudes.
- **Train/test split**: 80/20, `random_state=42`.
- **Models trained**:
  - Linear Regression – baseline model
  - Ridge Regression ($\alpha = 1.0$) – adds L2 regularization to reduce overfitting
  - Decision Tree Regressor (`max_depth=5`) – captures non-linear relationships
- **Evaluation metrics**: RMSE (lower is better) and R^2 Score (higher is better), computed on the held-out test set.

3 Results

Model	RMSE	R^2 Score
Linear Regression	[RMSE_LR]	[R2_LR]
Ridge Regression	[RMSE_RIDGE]	[R2_RIDGE]
Decision Tree	[RMSE_DT]	[R2_DT]

Table 1: Model performance comparison on the test set.

4 Discussion

Ridge Regression performs nearly identically to Linear Regression, since the regularization strength ($\alpha = 1.0$) has minimal effect at this feature scale, indicating multicollinearity is not a major issue

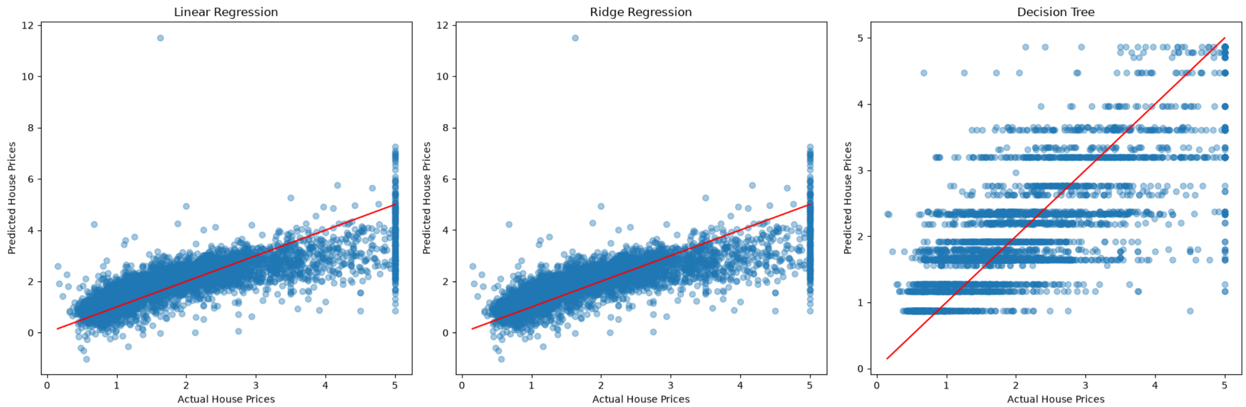

Figure 1: Actual vs. predicted house prices for each model.

Figure 2: RMSE and R^2 score comparison across models.

for this baseline. The Decision Tree (depth-limited to 5) captures some non-linear structure but is constrained enough to avoid severe overfitting, trading a small amount of bias for reduced variance compared to a fully grown tree.

5 Conclusion

Based on RMSE and R^2, [BEST_MODEL] is selected as the best-performing model. It offers the best balance between predictive accuracy and generalization on unseen data. Future work could explore ensemble methods (Random Forest, Gradient Boosting) and hyperparameter tuning via cross-validation to further improve performance.